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NUMERICAL LINEAR ALGEBRA · 1TD452

Signal and Image Processing

Mini-project 1: Direct Methods

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1 Introduction

Many problems in signal and image processing have the same basic form. We want to recover an unknown signal x , but we can only observe a distorted and noisy version of it,

$$b = Ax + e.$$

Here b is the observed data, A is the *forward operator* that describes how the measurement system (for example a blurring camera) distorts the signal, and e is unknown noise.

A natural first attempt is to solve the least squares problem $\min_x \|Ax - b\|_2^2$. But in many applications the matrix A is ill-conditioned. This means that small changes in b cause large changes on the solution.

What we often do is apply *regularization*. We add a penalty term that rewards solutions we consider reasonable. This gives us the regularized least squares problem

$$\min_x \|Ax - b\|_2^2 + \lambda^2 \|Lx\|_2^2. \quad (1)$$

The parameter $\lambda > 0$ controls the strength of the regularization. The matrix L encodes what we know about the signal in advance and depends on the application. For example, $L = I$ favours solutions of small size, while a difference operator such as

$$L = \begin{bmatrix} 1 & -1 & & & \\ & 1 & -1 & & \\ & & \ddots & \ddots & \\ & & & 1 & -1 \end{bmatrix} \in \mathbb{R}^{(n-1) \times n}$$

favours smooth solutions, since $(Lx)_i = x_i - x_{i+1}$ penalizes large jumps between neighbouring entries.

2 Method

2.1 Tikhonov regularization and SVD

Singular value decomposition (SVD) is a useful tool for understanding least-squares problems. We split A into simple building blocks.

$$A = U\Sigma V^T,$$

U and V are orthogonal matrices and Σ is a diagonal matrix holding the singular values $\sigma_1 \geq \sigma_2 \geq \dots \geq 0$ of A . We denote the columns of U and V by u_i and v_i .

Using SVD, the ordinary least squares solution of $\min_x \|Ax - b\|_2^2$ can be written as

$$x_{\text{LS}} = \sum_{i=1}^n \frac{u_i^T b}{\sigma_i} v_i.$$

Here we can see the problem. When A is ill-conditioned, some σ_i are tiny, and dividing by them blows up the noise in b . To prevent this, Tikhonov regularization can be used which adds a penalty on the size of x

$$\min_x f(x) := \|Ax - b\|_2^2 + \lambda^2 \|x\|_2^2,$$

which is Equation 1 with $L = I$. Expanding f gives

$$f(x) = (Ax - b)^T(Ax - b) + \lambda^2 x^T x = x^T A^T A x - 2x^T A^T b + b^T b + \lambda^2 x^T x.$$

Since f is strictly convex, x is the minimizer exactly when the gradient vanishes, $\nabla f(x) = 2(A^T A + \lambda^2 I)x - 2A^T b = 0$. This is the same as the *normal equations*

$$(A^T A + \lambda^2 I)x = A^T b.$$

The matrix $A^T A + \lambda^2 I$ is positive definite and hence invertible, so the normal equations have exactly one solution, $x_{\text{RLS}} = (A^T A + \lambda^2 I)^{-1} A^T b$.

To see what the regularization does, we insert the SVD into the normal equations and simplify using $U^T U = I$ and $V^T V = V V^T = I$:

$$\begin{aligned} (A^T A + \lambda^2 I)x_{\text{RLS}} &= A^T b \\ V \Sigma^T \cancel{U^T U} \Sigma V^T + \lambda^2 \cancel{V V^T} x_{\text{RLS}} &= V \Sigma^T U^T b \\ V(\Sigma^T \Sigma + \lambda^2 I)V^T x_{\text{RLS}} &= V \Sigma^T U^T b \\ \cancel{V^T V}(\Sigma^T \Sigma + \lambda^2 I)V^T x_{\text{RLS}} &= \cancel{V^T V} \Sigma^T U^T b \\ \underbrace{(\Sigma^T \Sigma + \lambda^2 I)}_{B} \underbrace{V^T x_{\text{RLS}}}_y &= \Sigma^T U^T b \\ B = \begin{bmatrix} \sigma_1^2 + \lambda^2 & & & \\ & \sigma_2^2 + \lambda^2 & & \\ & & \ddots & \\ & & & \sigma_n^2 + \lambda^2 \end{bmatrix} & \\ \Rightarrow y = B^{-1} \Sigma^T U^T b & \\ = \begin{bmatrix} \frac{1}{\sigma_1^2 + \lambda^2} & & & \\ & \frac{1}{\sigma_2^2 + \lambda^2} & & \\ & & \ddots & \\ & & & \frac{1}{\sigma_n^2 + \lambda^2} \end{bmatrix} \begin{bmatrix} \sigma_1 & & & \\ & \sigma_2 & & \\ & & \ddots & \\ & & & \sigma_n \end{bmatrix} \begin{bmatrix} u_1^T \\ u_2^T \\ \vdots \\ u_n^T \end{bmatrix} b & \end{aligned}$$

Because B^{-1} and Σ are both diagonal, each component of y can be read off directly:

$$y_i = \frac{\sigma_i}{\sigma_i^2 + \lambda^2} u_i^T b, \quad i = 1, \dots, n,$$

and since V is orthogonal we recover the solution as $x_{\text{RLS}} = Vy$.

We assume A has full rank, so every $\sigma_i > 0$. We can then split the coefficient and introduce the *filter factor* $\phi_i(\lambda)$:

$$\frac{\sigma_i}{\sigma_i^2 + \lambda^2} = \frac{1}{\sigma_i} \cdot \frac{\sigma_i^2}{\sigma_i^2 + \lambda^2} = \frac{\phi_i(\lambda)}{\sigma_i} \Rightarrow y_i = \frac{\phi_i(\lambda)}{\sigma_i} u_i^T b$$

Writing $x_{\text{RLS}} = Vy = \sum_i y_i v_i$ as a sum then gives

$$x_{\text{RLS}} = Vy = \sum_{i=1}^n \frac{\sigma_i}{\sigma_i^2 + \lambda^2} (u_i^T b) v_i = \sum_{i=1}^n \phi_i(\lambda) \frac{u_i^T b}{\sigma_i} v_i$$

Compared with x_{LS} , Tikhonov adds the filter factor $\phi_i(\lambda)$, which keeps terms with large singular values and suppresses those with small ones. The parameter λ sets the transition.

- if $\sigma_i \gg \lambda$, then $\phi_i(\lambda) \approx 1$ and the term is kept almost unchanged;
- if $\sigma_i = \lambda$, then $\phi_i(\lambda) = \frac{1}{2}$;
- if $\sigma_i \ll \lambda$, then $\phi_i(\lambda) \approx 0$ and the term is damped away.

2.2 Truncated SVD as a filter

The filter-factor form is not specific to Tikhonov regularization. Any method that damps each term of x_{LS} can be written as

$$x = \sum_{i=1}^n \phi_i \frac{u_i^T b}{\sigma_i} v_i,$$

and methods differ only in their choice of ϕ_i . Where Tikhonov damps the terms smoothly, the truncated SVD simply cuts them off. We pick a threshold $\delta > 0$ and let k be the number of singular values larger than δ . The filter factors are then

$$\phi_i = \begin{cases} 1 & \text{if } i \leq k \quad (\sigma_i > \delta) \\ 0 & \text{if } i > k \quad (\sigma_i \leq \delta) \end{cases}.$$

Substituting this in, only the first k terms survive:

$$x = \sum_{i=1}^k \frac{u_i^T b}{\sigma_i} v_i + \sum_{i=k+1}^n 0 \cdot \frac{u_i^T b}{\sigma_i} v_i = x_{\text{TLS}}.$$

The truncated least squares solution is thus a filter method with a sharp cut-off at $\sigma_i = \delta$, instead of the smooth transition of Tikhonov regularization.

2.3 Generalized SVD

So far we have used $L = I$, which only penalizes the size of x . For a general L the SVD of A is no longer enough. Instead we use the generalized SVD (GSVD), which decomposes both matrices at once:

$$A = U \Sigma_A W^{-1}, \quad L = V \Sigma_L W^{-1}. \quad (2)$$

Here U and V are orthogonal, W is invertible, and Σ_A and Σ_L are diagonal. The ratios $\gamma_i = \alpha_i / \beta_i$ are the *generalized singular values* of (A, L) .

2.4 General-form regularization and GSVD

We first characterize the solution of the general problem in Equation 1. Expanding its objective $f(x)$ as in the case $L = I$ but now with a general L gives

$$\begin{aligned} f(x) &= (Ax - b)^T (Ax - b) + \lambda^2 (Lx)^T (Lx) \\ &= x^T A^T A x - 2x^T A^T b + b^T b + \lambda^2 x^T L^T L x. \end{aligned}$$

Since f is convex, x is a minimizer exactly when $\nabla f(x) = 2(A^T A + \lambda^2 L^T L)x - 2A^T b = 0$. This is the normal equations for the general problem

$$(A^T A + \lambda^2 L^T L)x = A^T b. \quad (3)$$

The solution is unique when $A^T A + \lambda^2 L^T L$ is invertible. This is the case if $A^T A + \lambda^2 L^T L$ is positive definite, meaning, for $x \neq 0$,

$$x^T (A^T A + \lambda^2 L^T L)x = \|Ax\|^2 + \lambda^2 \|Lx\|^2 > 0$$

This implies that, when there is no unique solution, $Ax = Lx = 0$, which is true when x is in the nullspace of both A and L . So we require that A and L have no common nullspace, which is done by the choice of L .

Instead of solving Equation 3 directly, we use the GSVD Equation 2, which shows what the regularization does to each component. Here

$$\Sigma_A = \text{diag}(\alpha_1, \alpha_2, \dots, \alpha_r, 0), \quad \Sigma_L = \text{diag}(\beta_1, \beta_2, \dots, \beta_r, 0),$$

and we assume all $\alpha_i > 0$. Substituting the GSVD into the expanded $f(x)$ gives

$$\begin{aligned} f(x) = & x^T W^{-T} \Sigma_A^T U^T U \Sigma_A W^{-1} x - 2x^T W^{-T} \Sigma_A^T U^T b + b^T b \\ & + \lambda^2 x^T W^{-T} \Sigma_L^T V^T V \Sigma_L W^{-1} x. \end{aligned}$$

With the change of variables $y = W^{-1}x$ this becomes

$$f = y^T (\Sigma_A^T \Sigma_A + \lambda^2 \Sigma_L^T \Sigma_L) y - 2y^T \Sigma_A^T U^T b + b^T b.$$

Since W is invertible, minimizing over x is the same as minimizing over y . Setting the gradient with respect to y to zero gives

$$\begin{aligned} \nabla f(y) = & 2(\Sigma_A^T \Sigma_A + \lambda^2 \Sigma_L^T \Sigma_L)y - 2\Sigma_A^T U^T b = 0 \\ \Rightarrow & \underbrace{(\Sigma_A^T \Sigma_A + \lambda^2 \Sigma_L^T \Sigma_L)}_B y = \Sigma_A^T U^T b. \end{aligned}$$

$$\begin{aligned} B := \Sigma_A^T \Sigma_A + \lambda^2 \Sigma_L^T \Sigma_L &= \begin{bmatrix} \alpha_1^2 & & & \\ & \alpha_2^2 & & \\ & & \ddots & \\ & & & \alpha_r^2 \\ & & & & 0 \end{bmatrix} + \lambda^2 \begin{bmatrix} \beta_1^2 & & & \\ & \beta_2^2 & & \\ & & \ddots & \\ & & & \beta_r^2 \\ & & & & 0 \end{bmatrix} \\ &= \begin{bmatrix} \alpha_1^2 + \lambda^2 \beta_1^2 & & & \\ & \alpha_2^2 + \lambda^2 \beta_2^2 & & \\ & & \ddots & \\ & & & \alpha_r^2 + \lambda^2 \beta_r^2 \\ & & & & 0 \end{bmatrix} \end{aligned}$$

Solving for y gives

$$y = B^{-1} \Sigma_A^T U^T b$$

$$= \begin{bmatrix} \frac{1}{\alpha_1^2 + \lambda^2 \beta_1^2} & & & \\ & \frac{1}{\alpha_2^2 + \lambda^2 \beta_2^2} & & \\ & & \ddots & \\ & & & \frac{1}{\alpha_n^2 + \lambda^2 \beta_n^2} \end{bmatrix} \begin{bmatrix} \alpha_1 & & & \\ & \alpha_2 & & \\ & & \ddots & \\ & & & \alpha_n \end{bmatrix} \begin{bmatrix} u_1^T \\ u_2^T \\ \vdots \\ u_n^T \end{bmatrix} b$$

Because B^{-1} and Σ_A are both diagonal, each component of y can be read off directly:

$$y_i = \frac{\alpha_i}{\alpha_i^2 + \lambda^2 \beta_i^2} u_i^T b, \quad i = 1, \dots, n,$$

and we recover the solution as $x = Wy$.

As for Tikhonov, we split the coefficient and introduce the filter factor $\phi_i(\lambda)$

$$\frac{\alpha_i}{\alpha_i^2 + \lambda^2 \beta_i^2} = \frac{1}{\alpha_i} \cdot \frac{\alpha_i^2}{\alpha_i^2 + \lambda^2 \beta_i^2} = \frac{\phi_i(\lambda)}{\alpha_i}, \quad \phi_i(\lambda) = \frac{\alpha_i^2}{\alpha_i^2 + \lambda^2 \beta_i^2} = \frac{\gamma_i^2}{\gamma_i^2 + \lambda^2}.$$

Writing $x = Wy = \sum_i y_i w_i$, where w_i are the columns of W , as a sum then gives

$$x = \sum_{i=1}^n \frac{\alpha_i}{\alpha_i^2 + \lambda^2 \beta_i^2} (u_i^T b) w_i = \sum_{i=1}^n \phi_i(\lambda) \frac{u_i^T b}{\alpha_i} w_i.$$

Here $\gamma_i = \alpha_i / \beta_i$ compares how well A measures the component w_i with how much L penalizes it:

- $\gamma_i \gg \lambda$: $\phi_i(\lambda) \approx 1$, the term is kept;
- $\gamma_i \ll \lambda$: $\phi_i(\lambda) \approx 0$, the term is damped.

Since $\alpha_i^2 + \beta_i^2 = 1$, the damped terms have small α_i and large β_i . When $L = I$, only A decides what is damped. For a general L , α_i and β_i depend on both A and L . So by choosing L we decide which components are treated as unstable.

2.5 One-dimensional deblurring

We now apply the method to a concrete problem: removing blur from a one-dimensional signal. A blurred signal is modelled as

$$b(s) = \int_{-1}^1 K(s, t) x(t) dt + e(s), \quad K(s, t) = c e^{-\beta(s-t)^2},$$

where the Gaussian kernel K spreads each point of x over its neighbours, and a larger β gives a narrower blur. As true signal we use $x(t) = e^t \sin(\pi t)$. We use the box quadrature rule to turn the integral into a sum. We split $[-1, 1]$ into n boxes of width $h = 2/n$ with midpoints t_i . The integral is then approximated by the sum of the box areas,

$$b(t_i) \approx \sum_{j=1}^n hK(t_i, t_j)x(t_j) + e(t_i).$$

This is $b = Ax + e$ with $A_{ij} = hK(t_i, t_j)$, and we take the noise as $e = \delta \cdot \mathcal{N}(0, 1)$. We choose $c = \sqrt{\beta/\pi}$ so that the blur does not change the size of the signal, and use the first-difference matrix from the introduction as L .

The unregularized solution is computed with a standard least-squares solver. For the regularized solution we compute the GSVD of (A, L) once and use $x = Wy$ with $y_i = \alpha_i/(\alpha_i^2 + \lambda^2 \beta_i^2)u_i^T b$. To choose λ without knowing the true signal, we use the *discrepancy principle* [1]. The idea is that the solution should fit the data as well as the noise allows, but not better. So we pick the λ where the residual is as large as the noise, $\|Ax_\lambda - b\|_2 = \|e\|_2 \approx \delta\sqrt{n}$.

3 Results

3.1 Regularized vs. unregularized solution

Since $\text{cond}(A) \approx 10^{19}$, the unregularized solution is only amplified noise, with a relative error of about 10^{10} . The regularized solution follows the true signal well. Its error is 0.055, compared with 0.225 for the data b , see Figure 1.

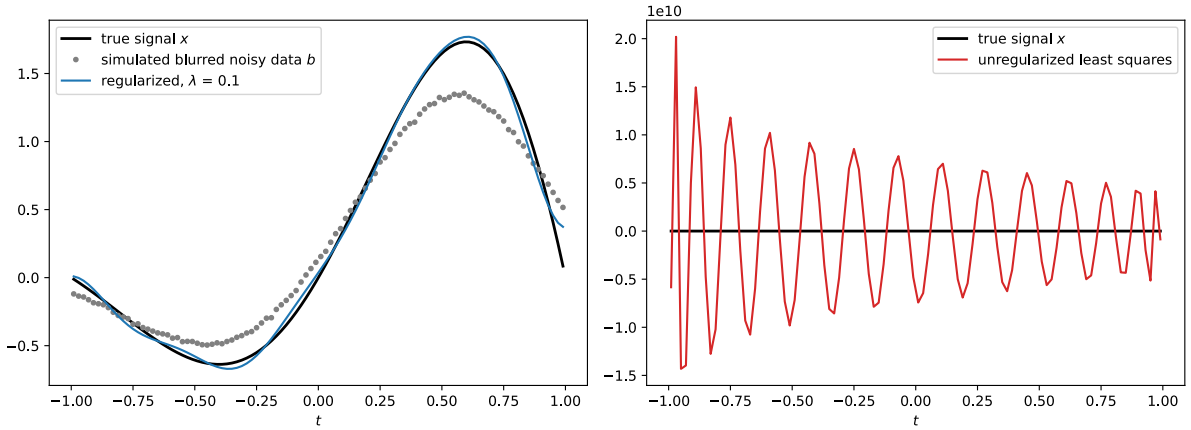


Figure 1: Regularized (left) and unregularized (right) solution.

3.2 Filter factors

Components with $\gamma_i \gg \lambda$ are kept and those with $\gamma_i \ll \lambda$ are suppressed. Increasing λ moves the cut-off to the right, so more components are removed, see Figure 2.

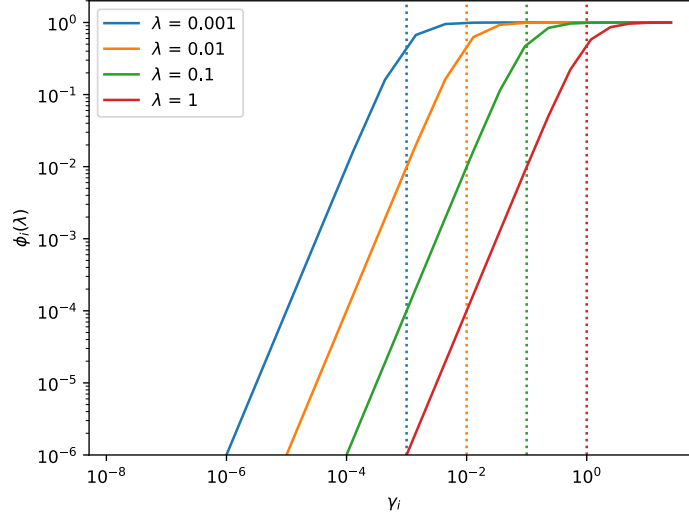


Figure 2: Filter factors for different λ . The dotted vertical lines show the point where $\gamma_i = \lambda$.

3.3 Effect of λ

Too small λ lets the noise through (under-regularization), too large λ smooths away the signal (over-regularization). The smallest error, 0.054, is at $\lambda \approx 0.093$, close to our hand-picked $\lambda = 0.1$, see Figure 3.

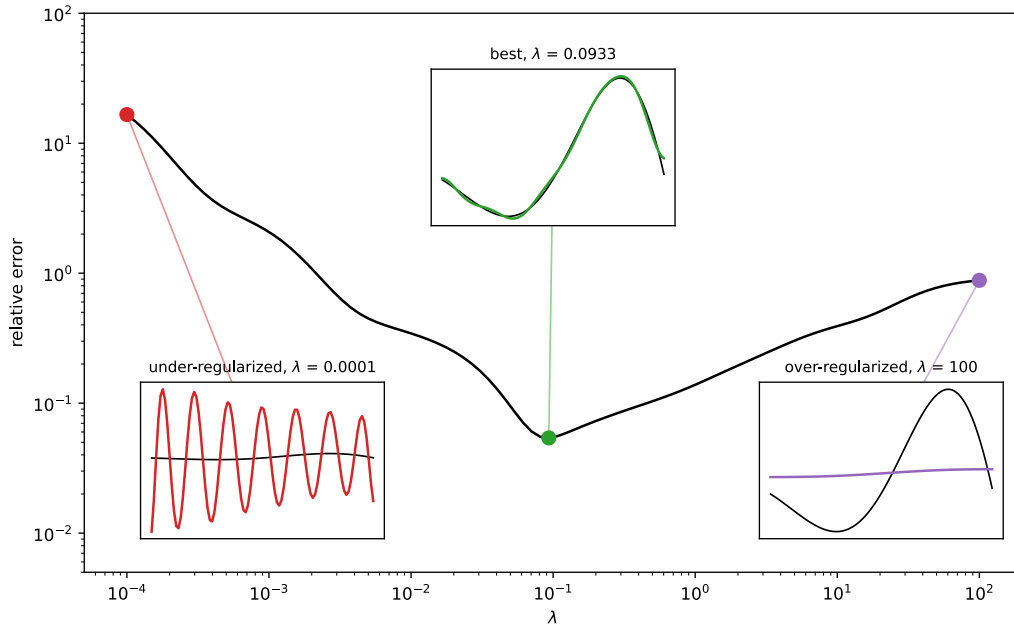


Figure 3: Relative error against λ .

3.4 Effect of the noise level

The unregularized error grows in proportion to δ , while the regularized error grows only slowly and the best λ increases with δ , see Table 1 and Figure 4.

δ	error, unregularized	best λ	error, regularized
10^{-3}	$7.1 \cdot 10^8$	0.027	0.040
10^{-2}	$7.1 \cdot 10^9$	0.093	0.054
10^{-1}	$7.1 \cdot 10^{10}$	0.66	0.116

Table 1: Errors for different noise levels.

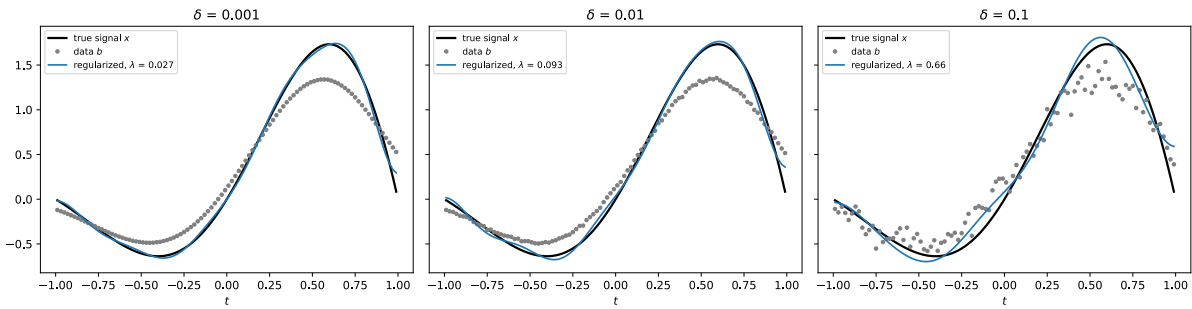


Figure 4: Regularized solutions for different noise levels.

3.5 Automatic choice of λ

The discrepancy principle gives a somewhat larger λ and a slightly larger error than the hand-picked and the best value, see Table 2.

method	λ	relative error
discrepancy principle	0.28	0.084
chosen by hand	0.1	0.055
best (from Figure 3)	0.093	0.054

Table 2: Choice of λ .

4 Discussion

Since $\text{cond}(A) \approx 10^{19}$, even small noise blows up without regularization. Regularization removes the components with small generalized singular values, where this happens. More noise needs a larger λ , so regularization becomes more important when the noise level grows.

In practice the true signal is unknown, so the best λ cannot be found directly. The discrepancy principle only needs the noise level and still gives a good reconstruction, although it slightly over-regularizes, which is typical for the method [1].

References

- [1] W. Karl, “3.6 - Regularization in Image Restoration and Reconstruction,” in *Handbook of Image and Video Processing (Second Edition)*, Second Edition., A. BOVIK, Ed., in Communications, Networking and Multimedia., Burlington: Academic Press, 2005, p. 183–V. doi: <https://doi.org/10.1016/B978-012119792-6/50075-9>.